FX Research Lab — research protocol report

Strategy 1.1 (strategy_v1_1.yaml) · EURUSD · 2020-01-01 → 2020-12-31 · generated 2026-11-05 09:29 UTC · EURUSD_2020.csv

Educational research tool. This software is an educational research and backtesting tool. It is not investment advice, not a trading signal service, and it has no broker connection or live execution. All results are simulated on historical data using the recorded strategy and execution assumptions. Simulated results are not evidence of a profitable edge and historical performance does not guarantee future results. The author is not a licensed or registered financial adviser. Trading leveraged foreign exchange involves a high risk of loss; nothing here should be used to place real trades.

Reproducibility

Field	Value
Dataset SHA-256	90d92b779e1744ddc9de7bcef4f73ab5f016b0ee3ed73128609b4a8ffb28b9a5
Configuration hash	9454655f074bee1463a4ec10d62ee672a3c1466ac0a974498bdc14ccf0c050b9
Execution model	first_executable_m1_price_after_bos
Spread sources	HISTORICAL_SPREAD, FIXED_CONFIGURED_SPREAD, NO_SPREAD_DATA
Slippage	0.1 pips
Timezone database	2026.5
Software	0.5.0 · source-sha256:29ff5ab67875481c0f492df189079cf3ca8bc6b0a52693c4da4e2aaad34bb945

Data

Rows	Gaps	Duplicates	Invalid	Weekend rows	Spread column	Source timezone
372275	1083	0	0	6165	no	embedded_offsets

■ Data gaps detected; no candles were fabricated.
■ Weekend-dated bars detected; verify source sessions and market closures.

Research protocol

Fixed-rule baseline evaluated on independent periods, rolling out-of-sample folds, a bootstrap of the trade sequence, one-at-a-time parameter variations and random entries that pay the same costs. Only out-of-sample figures may be quoted, always next to the random-benchmark percentile and the sensitivity table.

Baseline

Period	Trades	Expectancy	Total	Win rate	PF	Max DD	Sharpe (ann.)
full sample	15	-0.01R	-0.10R	40 %	0.99	+4.54R	-0.02

Independent chronological periods

Period	Trades	Expectancy	Total	Win rate	PF	Max DD	Sharpe (ann.)
train	8	+0.31R	+2.44R	50 %	2.05	+2.00R	0.95
validation	4	+0.11R	+0.46R	50 %	1.23	+2.00R	0.40
out_of_sample	3	-1.00R	-3.00R	0 %	0.00	+3.00R	-3.81

Out-of-sample metrics are the only ones that may be quoted as evidence; the baseline has no fitted parameters, so train/validation serve as independent replications.

Rolling out-of-sample (walk-forward)

9 folds · train 120 d, test 30 d. Concatenated OOS: 10 trades, -2.54R, expectancy -0.25R, 3/9 folds positive.

Period	Trades	Expectancy	Total	Win rate	PF	Max DD	Sharpe (ann.)
fold 1 OOS	1	+2.00R	+2.00R	100 %	—	+0.00R	3.38
fold 2 OOS	1	-1.00R	-1.00R	0 %	0.00	+1.00R	-3.46
fold 3 OOS	1	-1.00R	-1.00R	0 %	0.00	+1.00R	-3.31
fold 4 OOS	1	+0.46R	+0.46R	100 %	—	+0.00R	3.31
fold 5 OOS	2	-1.00R	-2.00R	0 %	0.00	+2.00R	-5.03
fold 6 OOS	1	+2.00R	+2.00R	100 %	—	+0.00R	3.38
fold 7 OOS	1	-1.00R	-1.00R	0 %	0.00	+1.00R	-3.31
fold 8 OOS	1	-1.00R	-1.00R	0 %	0.00	+1.00R	-3.38
fold 9 OOS	1	-1.00R	-1.00R	0 %	0.00	+1.00R	-7.94

Monte Carlo (trade-sequence bootstrap)

Paths	Trades / path	p(losing)	Final R p5	p50	p95	Max DD p50	Max DD p95
1000	15	52 %	-6.91R	-0.21R	+6.79R	+4.00R	+8.13R

Cost-matched random benchmark

Baseline expectancy -0.01R sits at percentile 51 of 200 simulations of random entries with identical stop distance, target, costs and session expiry (random median -0.02R/trade). Below the 95th percentile the result is indistinguishable from random timing.

Parameter sensitivity (one change at a time)

Parameter	Value	Trades	Expectancy	Total	Max DD
baseline	—	15	-0.01R	-0.10R	+4.54R
strategy:target.baseline	1.5R	15	-0.11R	-1.60R	+5.04R
strategy:target.baseline	2.0R	15	-0.01R	-0.10R	+4.54R
strategy:target.baseline	2.5R	15	-0.02R	-0.29R	+5.24R
strategy:target.baseline	3.0R	15	-0.03R	-0.48R	+5.24R
execution:slippage.baseline_pips	0.0	15	-0.00R	-0.04R	+4.53R

Parameter	Value	Trades	Expectancy	Total	Max DD
execution:slippage.baseline_pips	0.1	15	-0.01R	-0.10R	+4.54R
execution:slippage.baseline_pips	0.2	15	-0.01R	-0.16R	+4.56R
execution:slippage.baseline_pips	0.5	15	-0.08R	-1.15R	+5.42R
strategy:risk.baseline_equity_risk_fraction	0.0025	15	-0.01R	-0.10R	+4.54R
strategy:risk.baseline_equity_risk_fraction	0.005	15	-0.01R	-0.10R	+4.54R
strategy:risk.baseline_equity_risk_fraction	0.01	15	-0.01R	-0.10R	+4.54R
strategy:zones.expiry_m15_bars	4	14	-0.01R	-0.18R	+4.54R
strategy:zones.expiry_m15_bars	8	15	-0.01R	-0.10R	+4.54R
strategy:zones.expiry_m15_bars	12	15	-0.01R	-0.10R	+4.54R
strategy:setup.lifetime_m15_bars_from_sweep_confirmation	6	12	+0.01R	+0.14R	+3.54R
strategy:setup.lifetime_m15_bars_from_sweep_confirmation	8	15	-0.01R	-0.10R	+4.54R
strategy:setup.lifetime_m15_bars_from_sweep_confirmation	12	17	-0.30R	-5.03R	+7.54R
strategy:displacement.minimum_atr	0.75	15	-0.01R	-0.10R	+4.54R
strategy:displacement.minimum_atr	1.0	15	-0.01R	-0.10R	+4.54R
strategy:displacement.minimum_atr	1.25	15	-0.01R	-0.10R	+4.54R
strategy:stop.reference	sweep_extreme	16	-0.26R	-4.19R	+7.65R
strategy:stop.reference	zone_distal	15	-0.01R	-0.10R	+4.54R
strategy:stop.reference	leg1_origin	20	-0.34R	-6.82R	+11.49R
strategy:stop.max_distance_atr	3.0	15	-0.01R	-0.10R	+4.54R
strategy:stop.max_distance_atr	4.0	20	-0.16R	-3.11R	+5.68R
strategy:stop.max_distance_atr	5.0	22	-0.16R	-3.57R	+6.14R
strategy:liquidity.level_max_age_m15_bars	None	16	-0.25R	-3.95R	+8.54R
strategy:liquidity.level_max_age_m15_bars	960	12	-0.08R	-0.97R	+3.54R
strategy:liquidity.level_max_age_m15_bars	1920	15	-0.01R	-0.10R	+4.54R
strategy:liquidity.level_max_age_m15_bars	3840	16	-0.26R	-4.10R	+8.54R
multi:session	london_open_09:00-10:30	15	-0.01R	-0.10R	+4.54R
multi:session	new_york_open_15:00-16:30	30	-0.03R	-0.14R	+1.00R

One-at-a-time variations (and labelled multi-key session experiments); large swings between neighbouring values indicate parameter fragility. Experiments are not the baseline and must not be reported as such.

Limitations

- Simulated fills on historical M1 data; no live execution was involved.
- Costs follow config/execution.yaml; provider spreads are used when present.
- A positive result in one period is not evidence of a robust edge.
- Historical performance does not guarantee future results.

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